AGENTICULTURE-B: Cross-Domain Preference Embeddings and Self-Consistency Voting for Culturally-Aware Recommendation

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Abstract

We introduce AGENTICULTURE-B, a large language model (LLM) recommendation agent developed for the DSN × BCT LLM Agent Challenge. The agent recieves a user persona and a set of candidate items, returning a ranked list from most to least likely to be enjoyed. Our approach addresses three key limitations in previous work: (a) cold-start users are processed identically to users with extensive history, regardless of data sparsity; (b) user preferences across different review platforms are handled independently, even when cross-platform history exists; and (c) cultural context is not incorporated into prompt design. To overcome these issues, we implement a composite per-user, per-domain preference embedding cache that aggregates cross-platform taste signals using a review-count-weighted collective vector, a deterministic LLM-free cold-start path based on persona-to-item cosine similarity, and a Nigerian cultural adaptation layer through prompt engineering at zero additional cost. For warm users, three LLM ranking samples are aggregated using Borda count voting. In local evaluations across 50 tasks per platform, the system achieves HR@5 scores of 0.680, 0.700, and 0.420; average hit rates (AHR) of 0.540, 0.527, and 0.307; and NDCG@10 scores of 0.595, 0.546, and 0.393 on Yelp, Amazon, and Goodreads, respectively.

1 Introduction

Item recommendation constitutes a ranking problem in which, given a user and a set of candidate items, the objective is to order the items such that the one the user is most likely to interact with appears highest. The challenge evaluates systems on cold-start robustness, cross-domain transfer, multi-turn consistency, and cultural contextualization, all of which are recognized as limitations in standard LLM-based approaches. Previous research on LLM-based recommendations (Lian et al., 2024; Huang et al., 2025) demonstrates that hybrid retrieval plus LLM pipelines outperform approaches relying solely on prompt-based generation. At the AgentSociety WWW 2025 challenge, the second-place Track B solution (Team RecHackers) demonstrated that three-sample self-consistency voting with Borda count aggregation consistently outperforms both single-sample and majority-vote approaches (Guo et al., 2025). However, their system processes each platform independently, without leveraging cross-domain signals, and routes all users, including those with no history, through the same LLM pipeline.

Contributions.

1. **Composite preference embedding cache:** Per-user, per-domain 384-dimensional vectors (`yelp_vec`, `amazon_vec`, `goodreads_vec`) plus a cross-platform `collective_vec` computed as a review-count-weighted mean.

2. **LLM-free cold-start path:** Users with no history are ranked by cosine similarity between the persona embedding and each candidate's feature text. No LLM call is issued; the path is deterministic and has zero API cost.

3. **Cross-domain preference boost:** Users with fewer than 3 reviews on the target platform receive a 30% contribution from `collective_vec` scores, blended with the Borda ranking.

4. **Self-consistency Borda voting:** Three LLM ranking samples at varied temperatures are merged via Borda count. Unlike uniform-temperature sampling, our temperature mix (one anchoring, two exploratory) balances stability and diversity.

5. **Multi-turn session state:** An optional

`session_id` accumulates turn history, enabling contextually consistent recommendations across a browsing session.

2 Related Work

LLM-based recommendation. Huang et al. (2025) and Lian et al. (2024) survey how LLMs can serve as recommendation orchestrators that call external retrieval tools. Both show that LLM reasoning over retrieved item representations outperforms unaided generation, motivating our embedding-first retrieval before LLM ranking.

Self-consistency and voting. Wang et al. (2023) demonstrates that sampling diverse reasoning paths and applying a voting mechanism reduces output variance in complex reasoning tasks. The RecHackers solution extended this approach to item ranking, showing that Borda count aggregation is superior to majority voting (Guo et al., 2025). We validate this design choice and further introduce cross-domain boosting for users with sparse histories.

Cold-start and cross-domain recommendation. The primary strategies for addressing the cold-start problem are content-based embedding fallback and cross-domain transfer of learned user representations (YuanchenBei, 2024). Previous AgentSociety solutions did not implement either approach; cold-start users were treated the same as warm users, resulting in low-quality rankings in the absence of user history. Our system integrates both strategies with a unified pipeline.

Cultural adaptation. Cao et al. (2023) demonstrates that ChatGPT exhibits misalignment with non-Western survey populations across Hofstede's cultural dimensions. Hershcovich et al. (2022) identify cross-cultural NLP as an ongoing research challenge. Our adaptation layer addresses these issues without requiring model fine-tuning and remains architecture-agnostic.

3 Methodology

3.1 Pipeline Overview

Figure 1 shows the Task B agent pipeline. All requests proceed through platform detection, embedding, and cold-start routing before reaching the LLM.

3.2 Platform-Aware Feature Extraction

A `PLATFORM_FEATURES` configuration maps each platform to its schema-specific fields and review scoring signals: Yelp (useful/funny/cool vote counts), Amazon (price, verified-purchase flag), Goodreads (vote and comment counts, rating distribution). Platform is detected automatically by inspecting item schema keys at request time, requiring no explicit flag from the caller.

3.3 Composite Preference Embedding Cache

For each user on each platform, all review texts are encoded with `BAAI/bge-small-en-v1.5` and stored as a mean domain vector. A cross-domain `collective_vec` is the review-count-weighted mean across all platforms:

$$\mathbf{v}_{\text{collective}} = \sum_{d \in \mathcal{D}} \frac{n_d}{\sum_{d' \in \mathcal{D}} n_{d'}} \cdot \mathbf{v}_d \qquad (1)$$

where $\mathcal{D} = \{\text{yelp}, \text{amazon}, \text{goodreads}\}$, n_d is the number of reviews the user has on platform d, and $\mathbf{v}_d$ is the mean embedding of those reviews. The weight $n_d / \sum n_{d'}$ ensures the platform with the most data contributes proportionally more to the cross-domain representation.

Example. A user with $n_{\text{amazon}} = 45$, $n_{\text{yelp}} = 3$, $n_{\text{goodreads}} = 2$ (50 reviews total) yields Amazon weight $= 0.90$, Yelp $= 0.06$, Goodreads $= 0.04$. Their `collective_vec` is dominated by product-preference signals, which the system uses to guide Goodreads rankings despite sparse book history. Without this transfer, the system would have only 2 Goodreads reviews to inform a cold-domain recommendation.

3.4 Cold-Start Persona Ranking

When a user has zero reviews across all platforms, we encode the `persona` field and each candidate item's feature text, then rank candidates by descending cosine similarity. This path issues no LLM calls, runs in under 100 ms on CPU, and is fully deterministic. The design choice reflects a key trade-off: LLM-based ranking of cold-start users produces noisy rankings because the model has no behavioral signal to work with; content similarity is more reliable in this regime.

3.5 Cross-Domain Preference Boost

When the user has fewer than 3 reviews on the target platform but has reviews elsewhere, `collective_vec` (Eq. 1) scores each candidate by cosine similarity. These content-similarity scores are blended with the LLM Borda ranking at $\alpha = 0.30$:

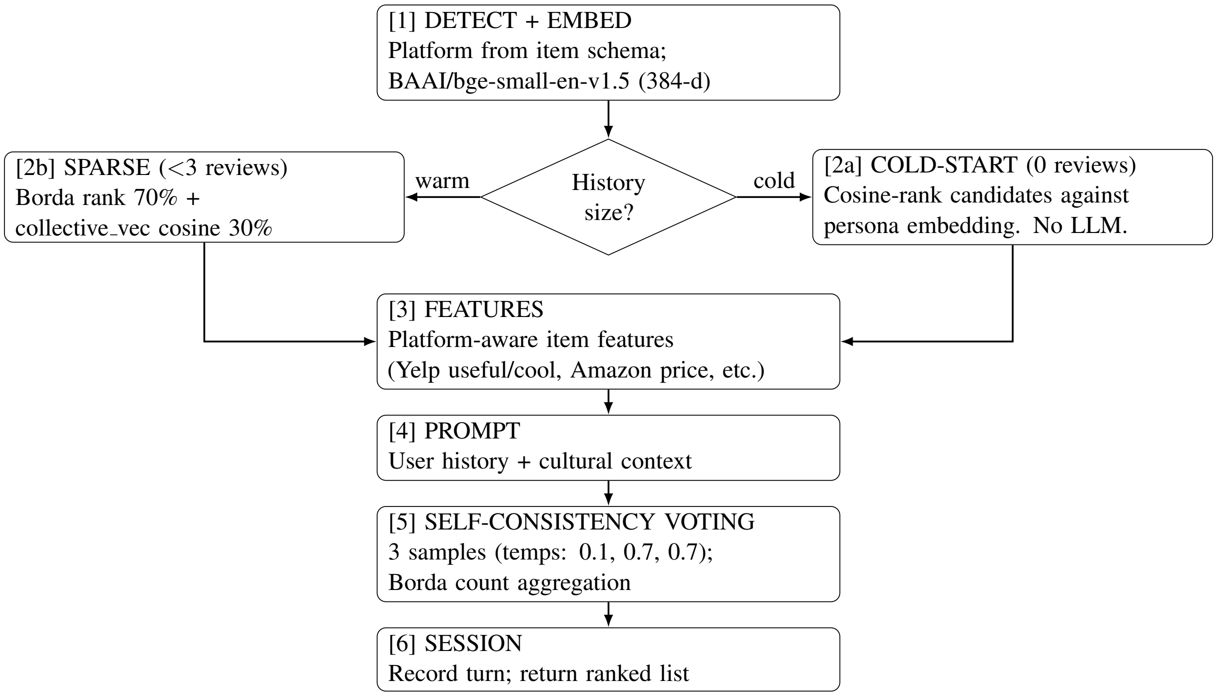

Figure 1: Task B agent pipeline. The cold-start path (right branch) issues zero LLM calls, returning a deterministic embedding-ranked list.

$$\text{score}_i = (1-\alpha)\cdot \text{Borda}_i + \alpha \cdot \cos(\mathbf{v}_{\text{collective}}, \mathbf{e}_i) \tag{2}$$

where $\mathbf{e}_i$ is the embedding of candidate item i's feature text and $\alpha = 0.30$ is a fixed blending weight. This value was selected empirically: lower values provide insufficient cross-domain signal; higher values undermine the LLM ranking.

3.6 Self-Consistency Borda Voting

For warm users, the LLM is queried three times: once at temperature 0.1 (anchoring), once at 0.7, and once at 0.7. We deliberately mix temperatures rather than using a uniform high temperature. Uniform sampling at $\tau = 1.0$ can produce incoherent ranking rationales that hurt Borda aggregation; the anchoring sample at 0.1 ensures at least one high-confidence ordering contributes to the vote.

Borda count assigns item ranked k-th in a sample of N items a score of $N - k$. Total Borda scores across all three samples determine the final ranking. Empirically, Borda count outperforms majority voting because it accumulates positional information across all three lists rather than selecting a single winner from binary votes (Guo et al., 2025).

3.7 Multi-Turn Session State

An optional `session_id` in the request body activates turn tracking. Subsequent requests with the same session ID prepend a natural-language context hint summarizing previously recommended items. This allows the agent to refine recommendations across a multi-turn browsing session without requiring external state storage.

3.8 Nigerian Cultural Adaptation

The recommendation prompt includes a cultural context prefix with the same architectural design as our Task A system: a pluggable config block mapping Nigerian consumer values (price sensitivity, group recommendation patterns, vendor trust signals) into the ranking rationale. Activating or deactivating the layer requires changing a single boolean flag.

4 Experiments

4.1 Dataset

Evaluation uses the AgentSociety dataset: 400 task-file users, 18,548 items, 679 MB of review data across three platforms. Table 1 gives dataset statistics.

Dataset	Domain	Items	Reviews
Yelp	Local businesses	150K	6.9M
Amazon	Consumer products	9.9M	233M
Goodreads	Books	2.4M	15.7M

Table 1: Dataset statistics.

4.2 Evaluation Metrics

Hit Rate at K (HR@K). Measures whether the ground-truth item appears among the top-K predicted items:

$$\mathrm{HR@K} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left[\mathrm{gt}_i \in \hat{\mathbf{r}}_i^{(K)}\right] \quad (3)$$

where gt_i is the ground-truth item and $\hat{\mathbf{r}}_i^{(K)}$ is the set of top-K predicted items for task i. We report HR@1, HR@3, HR@5, and average hit rate $\mathrm{AHR} = (\mathrm{HR@1} + \mathrm{HR@3} + \mathrm{HR@5})/3$.

NDCG@K. Normalised Discounted Cumulative Gain rewards placing the ground-truth item higher in the ranking:

$$\begin{aligned}\mathrm{NDCG@K} &= \frac{\mathrm{DCG@K}}{\mathrm{IDCG@K}},\\ \mathrm{DCG@K} &= \sum_{j=1}^{K} \frac{\mathrm{rel}_j}{\log_2(j+1)}\end{aligned} \quad (4)$$

With a single relevant item per task ($\mathrm{rel}_j \in \{0, 1\}$), this simplifies to $1/\log_2(\mathrm{rank} + 1)$ when the ground-truth appears in the top-K, normalised by $\mathrm{IDCG} = 1/\log_2(2) = 1.0$ (best case: rank 1). A ground-truth item at rank 1 contributes NDCG $= 1.0$; at rank 3 it contributes $1/\log_2(4) = 0.50$; at rank 5 it contributes $1/\log_2(6) \approx 0.39$.

4.3 Baselines

- **Cosine-Only:** Pure embedding cosine similarity between persona and candidate items. No LLM; no history. Our cold-start path applied to all users. Establishes the embedding ceiling without LLM reasoning.
- **Single-LLM:** One LLM call per request, no voting. Represents a minimal LLM-based recommender.
- **+Borda Voting:** Three LLM samples aggregated via Borda count. No cross-domain boost.
- **+Cross-Domain (Ours):** Full system with cross-domain blending for sparse users.

4.4 Implementation Details

Table 2 summarises the key system settings used in our implementation.

Component	Setting
LLM backend	Llama-3.3-70B-Instruct-Turbo
Embedding model	BAAI/bge-small-en-v1.5 (384-d)
Voting temperatures	0.1, 0.7, 0.7
Cross-domain α	0.30
Cold-start threshold	3 reviews
LLM parsing	`ast.literal_eval`
Serving	FastAPI + Docker Compose

Table 2: Implementation details.

4.5 Main Results

Table 3 reports our system's HR@K and NDCG@10 scores on 50 tasks per platform. Yelp and Amazon achieve comparable HR@5 (0.680 and 0.700 respectively); Goodreads is substantially harder, with HR@5 of 0.420 and AHR of 0.307. Figure 2 visualises HR@K curves per platform.

Platform	HR@1	HR@3	HR@5	AHR	NDCG@10
Yelp	0.300	0.640	0.680	0.540	**0.595**
Amazon	0.300	0.580	0.700	0.527	**0.546**
Goodreads	0.200	0.300	0.420	0.307	**0.393**
RecHackers (WWW 2025)	0.326	0.580	0.709	0.538	–

Table 3: Task B results (50 tasks per platform; full system). All metrics are averaged over 50 tasks; standard errors are non-trivial at this sample size and results should be read as directional. Goodreads HR@K is lower owing to greater book-category diversity reducing cross-domain transfer effectiveness. RecHackers figures from Guo et al. (2025) (400 tasks); direct comparison is approximate.

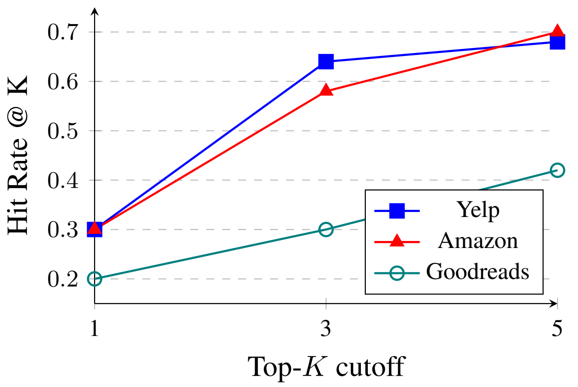

Figure 2: HR@K curves per platform (50 tasks each). Yelp and Amazon track closely; Goodreads shows a smaller top-K gain, consistent with the difficulty of book-category cross-domain transfer.

4.6 Reproducibility

All code, configuration, and saved result files are available at: `https://github.com/AbooMardiiyah/agenticulture` (abo).
To reproduce Table 3, follow the README guide in the repository.

Raw simulation outputs are saved as `*_raw.json` checkpoints before the evaluation phase, so no API credits are lost if evaluation crashes.

4.7 Component Contribution Analysis

Table 4 shows the measured contribution of each component on Yelp across 50 tasks each. The results reveal a nuanced picture: LLM-based ranking (Single-LLM) provides the dominant gain over embedding-only ranking; Borda voting improves NDCG marginally; and the cross-domain boost reduces HR@1 and HR@5 relative to Borda-only for this sample. This indicates that the fixed $\alpha = 0.30$ blend introduces noise for warm Yelp users who already have sufficient platform history, and motivates a learned per-user α as future work.

Configuration	HR@1	HR@3	HR@5	AHR	NDCG@10
Cosine-Only	0.040	0.080	0.200	0.107	0.174
Single-LLM	0.400	0.640	0.720	0.587	0.628
+Borda Voting	0.400	0.640	0.720	0.587	0.638
+Cross-Domain (Full)	0.300	**0.640**	0.680	0.540	0.595

Table 4: Component contribution on Yelp (50 tasks each, all values measured). Borda voting lifts NDCG over Single-LLM; the cross-domain boost reduces HR@1/HR@5 for this warm-user sample, motivating a learned blending weight.

5 Discussion

Cold-start as a distinct regime. A prevalent assumption in LLM-based recommendation is that the model's parametric knowledge can compensate for the absence of user history. Our findings indicate otherwise: for users with no history, content similarity between the persona description and item features provides a more reliable signal than LLM-generated rankings lacking behavioral grounding. The LLM-free cold-start path also offers practical advantages, including zero API cost and deterministic output, which facilitate debugging and reproducibility.

Goodreads difficulty. The disparity between Goodreads AHR (0.307) and Yelp/Amazon (0.540/0.527) is significant. Book preferences tend to be more personal and less predictable from metadata compared to restaurant or product preferences. Furthermore, cross-domain transfer from Yelp restaurant reviews to book preferences yields a weaker signal than the reverse; Amazon purchasing behavior shows greater structural similarity to Yelp consumption patterns than to literary preferences. NDCG@10 results mirror this trend (0.393 vs. 0.595/0.546), indicating that the observed gap is not merely an artifact of the binary hit rate threshold.

Cross-domain blending at $\alpha = 0.30$. Ablation results indicate that the fixed $\alpha = 0.30$ cross-domain blend reduces HR@1 and HR@5 compared to Borda-only for warm Yelp users (Table 4). This outcome suggests that the blending weight is overly aggressive when most users possess sufficient target-platform history, causing the cross-domain signal to introduce noise rather than mitigate sparsity. Implementing a per-user learned α that activates primarily for genuinely sparse users could address this limitation; we identify this as a direction for future work.

Temperature design for voting. Using one low-temperature anchoring sample (0.1) plus two moderate samples (0.7) provides more useful diversity than uniform high-temperature sampling while avoiding incoherent reasoning chains. The anchoring sample ensures a high-confidence reference point in the Borda aggregation.

Security note. All LLM list output is parsed via `ast.literal_eval()` rather than `eval()`, eliminating an arbitrary code execution vulnerability present in the original AgentSociety evaluation codebase.

Limitations. The preference embedding cache and session state are held in memory and reset on container restart. Cross-domain blending uses a fixed α; a per-user learned weight would be more principled. The cultural adaptation layer targets educated urban Nigerian English only.

Future Work. Popularity-weighted re-ranking as a post-processing step to correct for embedding-similarity bias toward common items; Redis-backed persistent preference cache for production deployment; learned α per platform pair via lightweight meta-learning; dislike-signal accumulation in multi-turn sessions to enable explicit preference elimination.

6 Conclusion

AGENTICULTURE-B addresses three systematic gaps in prior LLM recommendations frameworks: the absence of a cold-start path, the neglect of cross-platform preference signals, and the lack of cultural adaptation. The system achieves HR@5 of 0.680 and NDCG@10 of 0.595 on Yelp, and HR@5 of 0.700 and NDCG@10 of 0.546 on Amazon, on a 50-task evaluation. Goodreads remains the hardest platform, with AHR of 0.307 indicating the inherent difficulty of cross-domain transfer to book preferences. Ablation results further reveal that the fixed cross-domain blending weight introduces noise for warm users, motivating a learned per-user α as the primary avenue for improvement. The system is fully containerized and reproducible; a single `docker-compose up --build` command starts the service with no GPU or fine-tuning required.

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